# *Appendix C*
# Detail Display

Each Detail Display is designed as operator interface for one Controller or one major control unit. It displays and permits operation on variables, parameters and limits. The Detail Displays in the training applications use a similar layout and representation of information as with many industrial operator workstations.

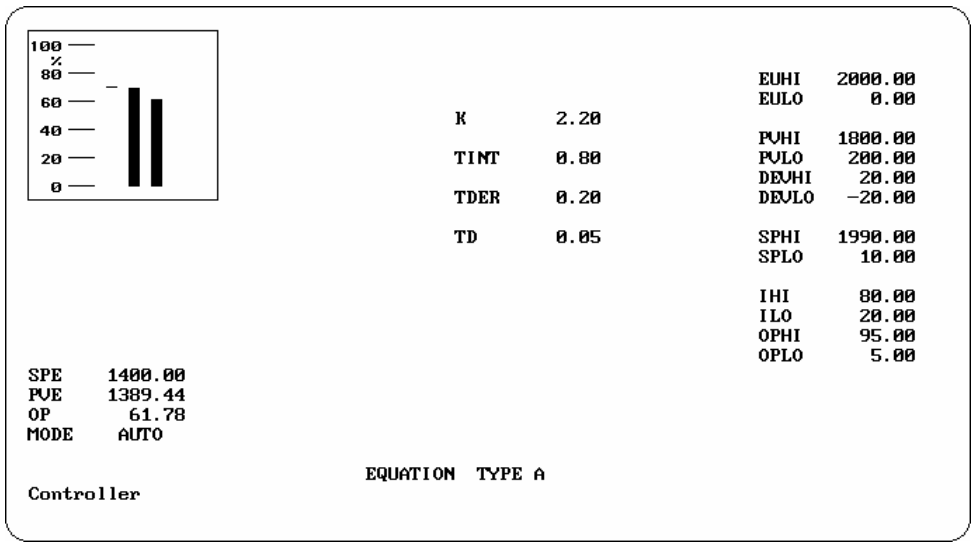


***Figure C.1***
***Detail Display***

The Detail Display is divided vertically in three parts, a left, middle and right section. The section on the left hand side is for **major control** variables and bar-graphs. The section in the middle is for **tuning constants**. The section on the right hand side is for **range, limit, alarm and status parameters**.

An explanation of the abbreviations (or acronyms) used in the manual and on the displays are explained in the sections below.

## Major Control Variables

**SPE** The Setpoint in Engineering Units or the target value for the controller. The objective is to control and keep the value of PVE at the value of SPE. 0% of range of the SPE is defined by EULO and 100% of range of SPE is defined by EUHI. A green marker, on the left side of bar-graphs represents the SPE in % of range.

**PVE** Process Variable in Engineering Units. The controller is to control and keep the PVE at the value of SPE. 0% of range of PVE is defined by EULO and 100% of range of PVE is defined by EUHI. A cyan bar-graph represents the PVE in % of range.

**OP** Output of controller in % to manipulate the Process or the setpoint of a secondary controller. A yellow bar-graph represents the OP in % of range. The human operator can change the value of OP in MANUAL mode only.

**MODE** Status variable to define the MODE of operation of a controller. MODE can assume the status MANUAL, AUTO, CASC, I-MAN, I-AUTO or I-CASC. MANUAL mode disables automatic control of the OP and enables the operator to manipulate the OP manually. AUTO mode enables automatic control of the OP and manual setting of SPE. CASC (cascade) mode enables automatic control, as in AUTO mode, but no operator setting of SPE is possible if another controller drives the SPE as its primary controller. If two conditions in a cascade control system are true, the primary controller assumes the MODE I-MAN, I-AUTO or I-CASC. These two conditions are:
the secondary controller is not in CASC mode and
the secondary controller is configured to initialize the primary controller.
In this situation, the secondary controller's SPE is used to set the primary controller's OP. The primary controller waits until the secondary controller has been switched into CASC mode. Then, the primary controller changes from I-MAN to MANUAL or from I-AUTO to AUTO or from I-CASC to CASC. The mode indicated after I, as in I-AUTO, is called pending mode. For example, I-AUTO means the primary controller is presently initialized with AUTO mode pending.

**Tag Name** A unique controller identification name. In the left bottom corner of Figure C1, we find the generic descriptor "Controller".

## Tuning Constants

**K** Controller gain. Generally, K is an overall controller gain for Proportional, Integral and Derivative control.

***Special Case:***
For K=0, Proportional and Derivative control are disabled. Integral control works with unit gain of 1.

**TINT** Time constant for Integral control (in minutes). The following example will give some idea of the meaning of TINT:
If the input of the integrator is a step of magnitude Y, the OP is a ramp function with a slope of K*Y / TINT.

**TDER** Time constant for Derivative control (in minutes). The control is based on "Rate of change of input" multiplied by K.

**TD** Time constant for Digital filtering of PVE. This is the time constant of a low-pass filter between the field input of a process variable (raw value of PV) and the PVE. The purpose is to filter out unwanted noise which has no bearing on the true behavior of PVE.

***Note:***
*Every field controller has an analogue input filter as well. This analogue filter has the purpose of filtering out all frequencies in an input signal that are too high for calculations within the controller's scan time. If these high frequencies have to be processed, a special controller with a very short scan time has to be selected.*

*This problem is referred to as aliasing (and states that the minimum sampling frequency must be at least twice the higher frequency component of the PV, or errors will occur in the representation of the digital data).*

## Range, Limit, Alarm and Status Parameters

**EUHI and EULO** High and low limits of the operational range of SPE and PVE in engineering units. Generally, a field input is transmitted to the controller as a 4 to 20mA (milli-Ampere) signal. The controller has to know the range of measurement. For example, 4mA (0% of range) may represent 0 C and 20mA (100% of range) may represent 2000 C.

**PVHI and PVLO** Alarm limits in engineering units, setting and displaying an alarm status within the controller. A PVHI alarm is set, if PVE is above PVHI. A PVLO alarm is set, if PVE is below PVLO. The value of PVE itself is not limited.

**DEVHI and DEVLO** Deviation Alarm limits in % of range, setting and displaying an alarm status within the controller when PVE deviates by too much from the SPE. A DEVHI alarm is set, if (PVE - SPE) is above DEVHI. A DEVLO alarm is set, if (PVE - SPE) is below DEVLO. The value of PVE itself is not limited.

**SPHI and SPLO** Set Point High or Set Point Low. This clamps the value of SPE at the limits SPHI and SPLO, when a primary controller attempts to drive SPE beyond SPHI or SPLO. Generally, industrial controllers permit the human operator to set SPE within full range, independently of SPHI and SPLO. SPHI and SPLO are in engineering units.

**IHI and ILO** Output limits for Integral control only (in %). An IHI alarm is set and the Integral calculation is suspended when both of the following are true; the value of OP is above IHI, and further integration would have increased OP (Integral wind-up high). An ILO alarm is set and Integral calculation is suspended when both of the following are true; the value of OP is below ILO, and further integration would have decreased OP (Integral wind-up low). Proportional and Derivative control is not affected at all by IHI or ILO.

**OPHI and OPLO** Output High and Output Low. OPHI and OPLO are in % and clamp the value the of the OP. OPHI and OPLO have priority over IHI and ILO. Generally, industrial controllers permit the human operator to set OP within full range, independent of OPHI and OPLO.

**CONFIG** Status variable, defining Initialization and / or PV-Tracking.

INIT initializes an upstream primary controller's OP if the initializing controller (secondary) is not in CASC mode.

TRACKING forces SPE to assume (track) the value of PVE if the controller is not in AUTO or CASC mode.

I & TR performs both Initialization and PV-Tracking.

**EQUATION** Status variable, defining different selections of inputs for PID control:

**TYPE A** calculates PID-Control on ERR (ERR = PVE-SPE).
**TYPE B** calculates PI-Control on ERR (ERR = PVE-SPE) and D-Control on PVE only.
**TYPE C** calculates I-Control on ERR (ERR = PVE-SPE) and PD-Control on PVE only.